ESE 3203 Question Map

Every question from the 2019, 2020, 2021, 2022, 2022B, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 3203 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×6, 2022B ×5, 2022 ×5, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). This course is almost all theory with diagrams, and the same questions repeat word for word: desalting and distillation (A3), reforming reactions (A4), delayed coking and FCC (A5), hydrotreating (A6), the amine process (B5), the LNG plant (B6) and separators (B3). Learn the memory tricks and practise each diagram once. The short numericals (API gravity, viscosity blend, VI, porosity, Darcy flow) are easy marks and each has a calculator.

Section A: Petroleum refining

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A1 Crude oil: what it is, composition, classification, impurities, why and how it is refined	Very likely 100%	Easy	19%
A6 Hydrotreating, hydrogen in the refinery, lube oil processing	Very likely 95%	Easy	30%

A2 Product properties and test numbers: octane and cetane number, API gravity, flash point, RVP, gasoline grades and pool, diesel classes	Very likely 86%	Easy	41%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A3 Distillation curves (TBP, ASTM, EFV), desalting, atmospheric and vacuum distillation	Very likely 100%	Medium	60%
A4 Catalytic reforming and isomerization	Very likely 100%	Medium	76%
✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)			
A5 Cracking and coking: thermal cracking, hydrocracking, delayed and fluid coking, visbreaking, FCC	Very likely 86%	Medium	89%

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
A7 Product blending: objectives, blending indices (RVP, viscosity, flash point), viscosity index numericals	Likely 77%	Medium	100%

Section B: Natural gas processing

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B5 Gas sweetening: reasons for removing H_2S and CO_2, amine process, iron sponge, molecular sieve, sulfur recovery	Very likely 95%	Easy	8%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B6 LNG and LPG: LNG plant and refrigeration cycles, storage and carriers, LPG uses and hazards, LPG vs LNG vs CNG, NGL recovery	Very likely 100%	Medium	32%
B1 Natural gas: what it is, origin, classification, condensate, reservoirs, Bangladesh outlook	Very likely 100%	Medium	52%
B3 Gas-liquid separators: vertical, horizontal, double-barrel, spherical, three-phase; internals; sizing	Very likely 100%	Medium	68%
B4 Water in natural gas: dehydration, water content (Campbell), hydrates	Very likely 100%	Medium	82%
✔ Pass line: 82% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.4 h)			

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B2 Gas properties and rock properties: z-factor, pseudo-reduced properties, density, B_g, compressibility, viscosity, porosity and permeability	Very likely 95%	Hard	92%

Block 4: likely · 0.8 h

Topic	Chance	Level	Covered
B7 Reservoir flow and gas measurement: Darcy's law (linear and radial), orifice meter, choosing a meter	Likely 73%	Medium	100%

Pass plan

Six papers: 2019, 2020, 2021, 2022, a second 2022 set marked "Rejected" in the file (shown here as **2022B**; it repeats most 2022 topics, so it is a good extra practice paper) and 2023. Normal papers: 210 marks, answer any THREE of four questions in each section (35 marks each). 2020 was a short exam (120 marks, 1.5 h, any TWO of three per section). Section A is petroleum refining (Shameem sir) and Section B natural gas processing (Muhit sir); **in 2021 the sections were swapped** (Section A natural gas, Section B refining), so this pack sorts by **topic**, not question number. **Minimum pass set:** refining A1-A7 (crude composition, product properties, distillation and desalting, reforming, cracking, hydrotreating and lube, blending numericals) and gas B1-B6 (natural gas and condensate, gas properties, separators, dehydration and hydrates, sweetening, LNG/LPG); B7 (Darcy flow, orifice) is the next to add. In the paper: Section A answer Q1, Q2 and Q3 or Q4; Section B answer Q5, Q6 and Q7 (Q8 if LPG/LNG suits you). Answers follow the teachers' slides first (Shameem sir: refining; Muhit sir: natural gas L1-L17) and the books they use: Rao *Modern Petroleum Refining Processes* and Fahim et al. for refining, Sanjay Kumar *Gas Production Engineering*, Guo & Ghalambor *Natural Gas Engineering Handbook* and Ahmed for gas. Each answer says where a textbook was used. Every numerical was re-checked in Python.

SECTION A: PETROLEUM REFINING

7 topics

SECTION B: NATURAL GAS PROCESSING

6 topics

cover 100% of all Section A marks asked

- Crude oil composition, impurities, classification, why refine (A1)
- Octane and cetane numbers, API numerical, gasoline pool (A2)
- Distillation curves, desalting, ADU/VDU (A3); catalytic reforming and isomerization (A4)
- Cracking: delayed coking, visbreaking, FCC (A5); hydrotreating and lube oil (A6); blending numericals (A7)

cover 92% of all Section B marks asked

- Natural gas, condensate separation, origin (B1); gas properties and z-factor numericals (B2)
- Separators and internals, sizing numerical (B3)
- Dehydration, Campbell water content, hydrates (B4); sweetening (B5)
- LNG and LPG (B6); backup: Darcy flow and orifice meter (B7)

- **Refining Q1 every year (crude and products):** crude composition and impurities, classification, why refine; octane/cetane definitions; one API numerical (A1, A2).
- **Refining Q2 every year (separation and reforming):** TBP/ASTM/EFV curves, desalting with diagram, ADU/VDU tower arrangement, catalytic reforming reactions (A3, A4).
- **Refining Q3-Q4 (conversion and finishing):** delayed coking diagram, visbreaking coil vs soaker, FCC diagram, hydrotreating diagram, lube oil steps, blending objectives and the viscosity-blend numerical (A5-A7).
- **Gas Q5-Q6 (gas and separators):** condensate separation flow diagram, origin of gas, gas properties (B_g, c_g, z), separators with internals (de-foaming plates, vortex breaker), separator sizing (B1-B3).
- **Gas Q7-Q8 (treatment and products):** dehydration reasons, Campbell water content numerical (charts given), hydrates, amine sweetening, H_2S/CO_2 reasons, LNG plant schematic, LPG uses and hazards, LPG/LNG/CNG table (B4-B6); Darcy linear and radial numericals (B7).
- **Always draw:** refinery block scheme, desalter, ADU + VDU, SR reformer with temperature profile, delayed coker, FCC, hydrotreater, separators (vertical, horizontal, three-phase), hydrate curve, glycol and amine units, LNG plant block diagram.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: PETROLEUM REFINING: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2022B	2023	Years	Marks
A1 Crude oil: what it is, composition, classification, impurities, why and how it is refined	24	20	50	17	16	13	6/6	140
A6 Hydrotreating, hydrogen in the refinery, lube oil processing		20	8	17	12	20	5/6	77
A2 Product properties and test numbers: octane and cetane number, API gravity, flash point, RVP, gasoline grades and pool, diesel classes	20	16		18	19	14	5/6	87
A3 Distillation curves (TBP, ASTM, EFV), desalting, atmospheric and vacuum distillation	12	14	15	35	28	35	6/6	139
A4 Catalytic reforming and isomerization	58	10	12	8	20	8	6/6	116
A5 Cracking and coking: thermal cracking, hydrocracking, delayed and fluid coking, visbreaking, FCC			8	27	27	35	4/6	97
A7 Product blending: objectives, blending indices (RVP, viscosity, flash point), viscosity index numericals	26			18	18	15	4/6	77

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: NATURAL GAS PROCESSING: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2022B	2023	Years	Marks

B5 Gas sweetening: reasons for removing H_2S and CO_2, amine process, iron sponge, molecular sieve, sulfur recovery		10	20	12	8	20	5/6	70
B6 LNG and LPG: LNG plant and refrigeration cycles, storage and carriers, LPG uses and hazards, LPG vs LNG vs CNG, NGL recovery	78	5	47	30	22	20	6/6	202
B1 Natural gas: what it is, origin, classification, condensate, reservoirs, Bangladesh outlook	27	34	35	25	20	25	6/6	166
B3 Gas-liquid separators: vertical, horizontal, double-barrel, spherical, three-phase; internals; sizing	15	10	25	25	25	35	6/6	135
B4 Water in natural gas: dehydration, water content (Campbell), hydrates	20	21	25	18	23	15	6/6	122
B2 Gas properties and rock properties: z-factor, pseudo-reduced properties, density, B_g, compressibility, viscosity, porosity and permeability		20	35	10	10	10	5/6	85
B7 Reservoir flow and gas measurement: Darcy's law (linear and radial), orifice meter, choosing a meter				20	32	15	3/6	67

Section A: Petroleum refining

topics A1-A7 (usually questions 1-4; questions 5-8 in 2021)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A1 Crude oil: what it is, composition, classification, impurities,

Pass set · Very likely · 100% · Easy

why and how it is refined

Asked in 6 of 6 papers · 140 marks in total

STUDY THESE

- Crude oil = naturally occurring mixture of hydrocarbons (C 83-87%, H 10-14%) with S, N, O compounds, metals and salts; a fossil fuel formed from buried marine organisms under heat and pressure
- Hydrocarbon types: paraffins C_nH_{2n+2} (normal and iso), naphthenes C_nH_{2n} (rings), aromatics (benzene rings), olefins (made in processing)
- Classification: paraffinic, naphthenic, aromatic, asphaltic/mixed base; by API (light > 31.1, medium, heavy < 22.3, bitumen < 10); by sulfur (sweet < 0.5%, sour); U.S. Bureau of Mines key fractions; K_W
- Refining: separation (distillation), conversion (cracking, reforming, alkylation, isomerization), treatment (desalting, hydrotreating), blending

> 2023 Q1(a) 6 marks
>
> What is crude oil? Explain its classification based on composition.

> 2023 Q1(c) 7 marks
>
> What are the key properties and characteristics of petroleum that need to be evaluated?

> 2022B Q1(a) 10 marks
>
> Briefly discuss the composition of petroleum with figures of commonly found hydrocarbon compounds in crude oil.

> 2022B Q1(b) 6 marks
>
> Why does crude oil require refining, and what are the essential processes involved?

> 2022 Q1(a) 10 marks
>
> Briefly describe crude oil compositions. What are the impurities associated with a crude oil?

> 2022 Q1(c) 7 marks
>
> Explain briefly why crude oil needs to be refined and how this is done.

> 2021 Q5(a) 15 marks
>
> Why is petroleum called a “fossil fuel”? Briefly discuss the composition of petroleum with figures of commonly found hydrocarbon compounds.

> 2021 Q5(b) 8 marks

> What is meant by “heat capacity”? Why is dehydration needed in crude oil refining?

> 2021 Q5(c) 12 marks
>
> Adverse effects of sulfur compounds in refineries; test methods for sulfur; types of stabilizers for crude oil stabilization.

> 2021 Q8(c) 15 marks
>
> Describe the function of an energy engineer in a petroleum refinery.

> 2020 Q1(a) 10 marks
>
> Classify and briefly describe the basic types of processes and operations in petroleum refining.

> 2020 Q1(b) 10 marks
>
> Explain: BPSD, BPCD, refinery complexity, °API, K_W.

> 2019 Q1(a) 15 marks
>
> Briefly describe crude oil compositions. What are the impurities associated with crude oil? Give examples of each.

> 2019 Q3(a) 9 marks
>
> Describe in brief: total acid number, TBP distillation, natural gas condensate.

Full answer: ESE 3203 Study Pack, topic A1.

A6 Hydrotreating, hydrogen in the refinery, lube oil processing

Pass set · Very likely · 95% · Easy

Asked in 5 of 6 papers · 77 marks in total

STUDY THESE

- Hydrotreating: feed + H_2 over Co-Mo or Ni-Mo on alumina at 290-400 °C, 10-100 bar; removes S (HDS), N (HDN), O, metals, saturates olefins; H_2S removed by amine; protects reformer/FCC catalysts and meets fuel specs
- Hydrogen sources: catalytic reformer (by-product), steam methane reforming ($CH_4 + H_2O \rightarrow CO + 3H_2$, shift $CO + H_2O \rightarrow CO_2 + H_2$), partial oxidation, recovery from off-gases
- Lube oil: vacuum distillation → propane deasphalting → solvent extraction of aromatics (furfural, phenol, NMP) → solvent dewaxing (MEK-toluene) → hydrofinishing → blending with additives

- Hydrofinishing: mild hydrotreating of the dewaxed base oil to improve colour, odour, oxidation stability

> 2023 Q4(d) 12 marks
>
> Explain the hydrotreating process with a neat labeled diagram.

> 2023 Q4(b) 8 marks
>
> Describe the hydrofinishing process in lube oil processing with a labeled diagram.

> 2022B Q4(c) 12 marks
>
> Same as 2023 4(d).

> 2022 Q4(d) 9 marks
>
> Same as 2023 4(d).

> 2022 Q4(b) 8 marks
>
> Detailed description of lube oil processing with a diagram.

> 2021 Q8(a) 8 marks
>
> Clarify the necessity of using lube oil in vehicles.

> 2020 Q3(b) 10 marks
>
> With a neat sketch explain the hydrotreating process.

> 2020 Q3(c) 10 marks
>
> Why is hydrogen required in a refinery? From which sources can it be obtained?

Full answer: ESE 3203 Study Pack, topic A6.

A2 Product properties and test numbers: octane and cetane number, API gravity, flash point, RVP, gasoline grades and pool, diesel classes

Pass set · Very likely · 86% · Easy

Asked in 5 of 6 papers · 87 marks in total

STUDY THESE

- Octane number = % iso-octane in an iso-octane/n-heptane mix that knocks like the fuel (knock resistance, SI engines); RON (600 rpm, city), MON (900 rpm, highway), sensitivity = RON − MON, PON = (RON + MON)/2

- Cetane number = % n-cetane in a cetane/α-methylnaphthalene mix with the same ignition delay (CI engines, 40-55 for diesel)
- API = 141.5/SG − 131.5 (SG at 60 °F); water = 10; flash point = lowest temperature at which vapours flash with a flame; RVP = vapour pressure at 37.8 °C (100 °F) in the Reid bomb
- Gasoline grades: regular 85-88, mid-grade 88-90, premium > 90; gasoline pool: LSR, isomerate, reformate, FCC, hydrocracked, alkylate, polymer gasoline, n-butane, oxygenates (MTBE, ETBE, TAME, ethanol)

> `2023 Q1(b) 6 marks`
>
> What is the octane number, and how is it used to evaluate the anti-knock performance of gasoline?

> `2023 Q1(e) 8 marks`
>
> Define: cetane number, API gravity, flash point, RVP.

> `2022B Q1(c) 7 marks`
>
> What is the anti-knock performance of gasoline, and how does it relate to octane numbers?

> `2022B Q1(d) 7 marks`
>
> What is the cetane number, and how is it used to evaluate diesel fuel quality?

> `2022B Q1(e) 5 marks`
>
> The specific gravity of crude oil is 0.821; determine its API gravity.

> `2022 Q1(b) 7 marks`
>
> Same question as 2023 1(b).

> `2022 Q1(e) 6 marks`
>
> Define: octane number, API gravity, flash point.

> `2022 Q1(d) 5 marks`
>
> The specific gravity of NGC is 0.802; determine its API gravity.

> `2020 Q2(a) 6 marks`
>
> Classify different grades of gasoline based on octane number.

> `2020 Q2(b) 10 marks`
>
> Components of the gasoline pool? Why are oxygenates added to gasoline? Name a few oxygenates.

> `2019 Q1(c) 7 marks`

> What is octane number? Explain the effect of variables on an engine's octane rating.

> 2019 Q1(b) 8 marks
>
> Classify petroleum products based on boiling point and carbon numbers.

> 2019 Q1(d) 5 marks
>
> Classify diesel fuel in terms of engine speed.

Full answer: ESE 3203 Study Pack, topic A2.

A3 Distillation curves (TBP, ASTM, EFV), desalting, atmospheric and vacuum distillation

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 139 marks in total

STUDY THESE

- TBP: batch, 15 theoretical plates, reflux 5 (15/5), atmospheric then 40 mmHg; ASTM D86/D1160: no plates, no reflux (Engler flask), for products; EFV: continuous flash, vapour stays with liquid, for designers; sharpness TBP > ASTM > EFV
- Desalting: 3-10 vol% wash water + demulsifier, heat to 90-150 °C, electrostatic coalescence (secondary field ~1000 V/cm, primary ~600 V/cm), settling (Stokes); needed if salt > 10 lb/1000 bbl
- ADU: crude preheated to ~280 °C, furnace to 350-400 °C, 30-50 trays, side strippers, pump-around; products LPG, naphtha, kerosene, diesel/gas oil, atmospheric residue
- VDU: residue heated to 390-450 °C, 25-40 mmHg flash zone, steam ejectors, packing, wide column; products LVGO, HVGO, lube cuts, vacuum residue

> 2023 Q1(d) 8 marks
>
> What are distillation curves? How does ASTM differ from EFV?

> 2023 Q2(a) 5 marks
>
> What is meant by True Boiling Point (TBP) distillation?

> 2023 Q2(b) 10 marks
>
> Describe the dehydration and desalting of crude oil with a diagram.

> 2023 Q2(c) 12 marks
>
> Explain atmospheric and vacuum distillation of crude oil showing the arrangement of towers to obtain principal fractions.

> 2022B Q3(a) 8 marks
>
> Describe in brief: TBP distillation, ASTM distillation, API gravity.

> 2022B Q2(a) 10 marks
>
> Describe crude oil desalting with a diagram. Operating variables of the desalter?

> 2022B Q2(b) 10 marks
>
> What are ADU and VDU? Schematic of a distillation tower with products and their temperature ranges.

> 2022 Q2(d) 7 marks
>
> What do you understand by TBP distillation? Explain.

> 2022 Q3(a) 8 marks
>
> Discuss ASTM distillation characteristics with a neat diagram.

> 2022 Q2(a) 10 marks
>
> Describe the crude oil desalting process with a diagram.

> 2022 Q2(b) 10 marks
>
> Same as 2023 2(c).

> 2021 Q6(a) 15 marks
>
> Briefly explain the crude oil distillation process.

> 2020 Q2(c) 14 marks
>
> Explain vacuum distillation with a flow diagram.

> 2019 Q4(d) 12 marks
>
> Same as 2022B 2(a).

Full answer: ESE 3203 Study Pack, topic A3.

A4 Catalytic reforming and isomerization

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 116 marks in total

STUDY THESE

- Reforming upgrades heavy naphtha (RON 50-60) to reformate (RON 90-105), makes aromatics and H_2; Pt (0.2-0.6%) on chlorided alumina, bifunctional (metal: dehydrogenation; acid: isomerization, cyclization)
- Reactions: dehydrogenation of naphthenes (fast, very endothermic), dehydroisomerization of alkylcyclopentanes, dehydrocyclization of paraffins

(slow), isomerization (mild exothermic), hydrocracking (slow, exothermic, unwanted)

- SR (semi-regenerative): 3-4 fixed beds with interheaters, 12-15 bar, regenerated every 6-24 months; CCR: moving bed, 3-10 bar, continuous regeneration, Pt-Sn, RON 98-105+
- Isomerization: n-C5/n-C6 → branched isomers at 95-205 °C, Pt/chlorided alumina or zeolite; raises RON by 12-20; catalyst poisons: S, N, water, metals, coke

> `2023` `Q2(d)` `8 marks`
>
> What is catalytic reforming? Briefly describe the major reactions.

> `2022B` `Q2(c)` `9 marks`
>
> Same as 2023 2(d).

> `2022B` `Q2(d)` `6 marks`
>
> Distinguish between CCR and SR reforming.

> `2022B` `Q4(d)` `5 marks`
>
> Main reactions of an isomerization process.

> `2022` `Q2(c)` `8 marks`
>
> Same as 2023 2(d).

> `2021` `Q6(c)` `12 marks`
>
> Write down the catalytic reforming reactions.

> `2020` `Q3(a)` `10 marks`
>
> What is catalyst poisoning? Examples of catalyst poisons.

> `2019` `Q2(a)` `10 marks`
>
> Same as 2023 2(d).

> `2019` `Q2(b)` `12 marks`
>
> Process diagram of the SR reforming process; variation of temperature and effluent composition in the reactors.

> `2019` `Q2(c)` `8 marks`
>
> Distinguish between CCR and SR reforming.

> `2019` `Q2(d)` `5 marks`
>
> Write down C_5 and C_6 isomers and their RON values.

> `2019` `Q4(a)` `5 marks`
>
> What are the objectives of light naphtha isomerization?

> `2019` `Q4(b)` `10 marks`

Main and side reactions of an isomerization process.

2019 Q4(c) 8 marks

Light naphtha (SG 0.724, 1000 m^3/h) to isomerization; converted yields C_3 0.348, iC_4 0.619, nC_4 1.770, C_5^+ 97.261 wt%: product composition.

Full answer: ESE 3203 Study Pack, topic A4.

A5 Cracking and coking: thermal cracking, hydrocracking, delayed and fluid coking, visbreaking, FCC

Pass set · Very likely · 86% · Medium

Asked in 4 of 6 papers · 97 marks in total

STUDY THESE

- Thermal cracking: heat (450-750 °C) and pressure break large molecules, free radical, no catalyst; hydrocracking: catalytic cracking + hydrogenation at high H_2 pressure, bifunctional catalyst, saturated, sulfur-free products
- Delayed coking: residue heated to ~500 °C in a furnace, cracks in large coke drums (two drums, switched every ~24 h); vapours to fractionator; coke cut out by water jets (2000-4500 psig)
- Visbreaking: mild thermal cracking of vacuum residue to cut viscosity and pour point; coil (474-500 °C in the furnace, quench at exit) vs soaker (427-438 °C, extra time in a soaking drum)
- FCC: hot catalyst (zeolite) meets feed in the riser (480-540 °C), cracks in seconds; spent (coked) catalyst steam-stripped and regenerated by burning coke (650-815 °C); heat balance

2023 Q3(a) 6 marks

Distinguish between thermal cracking and catalytic hydrocracking.

2023 Q3(b) 10 marks

Describe the delayed coking process with a neat sketch.

2023 Q3(c) 8 marks

Define visbreaking. Principal reactions during visbreaking.

2023 Q3(d) 11 marks

With neat labeled diagrams explain the working of an FCC unit.

2022B Q3(b) 10 marks

> Same as 2023 3(b).

> 2022B Q3(c) 8 marks
> Same as 2022 3(c).

> 2022B Q3(d) 9 marks
> What distinguishes fluidized-bed catalytic cracking from fixed-bed catalytic cracking?

> 2022 Q3(b) 10 marks
> Same as 2023 3(b).

> 2022 Q3(c) 7 marks
> What is visbreaking? Differentiate between coil and soaker visbreaker.

> 2022 Q3(d) 10 marks
> Same as 2023 3(d).

> 2021 Q6(b) 8 marks
> Basic functions and types of catalytic cracking. What is “spent catalyst”?

Full answer: ESE 3203 Study Pack, topic A5.

A7 Product blending: objectives, blending indices (RVP, viscosity, flash point), viscosity index numericals

Pass set · Likely · 77% · Medium

Asked in 4 of 6 papers · 77 marks in total

STUDY THESE

- Objective: combine semi-finished streams to meet product specifications at the lowest cost and maximum profit
- Linear (additive) properties: SG, sulfur, boiling point: $P_{blend} = \Sigma x_i P_i$; non-linear ones use blending indices: RVP BI = $RVP^{1.25}$
- Viscosity: BI = logv/(3 + logv); $BI_{blend} = \Sigma x_{v,i} BI_i$; $v_{blend} = 10^{3BI/(1-BI)}$
- Viscosity index VI = 100(L − U)/(L − H): U = oil’s viscosity at 40 °C, L and H = 40 °C viscosities of VI 0 and VI 100 reference oils with the same 100 °C viscosity

> 2023 Q4(a) 7 marks
> Objectives of product blending? How to calculate viscosity blending.

> 2023 Q4(c) 8 marks
>
> What is viscosity index? Calculate VI: ν_{40} = 73.3 cSt, ν_{100} = 8.86 cSt, L = 120, H = 70.

> 2022B Q4(a) 8 marks
>
> Same as 2022 4(a).

> 2022B Q4(b) 10 marks
>
> Same with 75, 100 and 200 cSt.

> 2022 Q4(a) 8 marks
>
> Objectives of product blending? How to calculate RVP blending.

> 2022 Q4(c) 10 marks
>
> Estimate the viscosity of a blend: 2000 bbl at 72 cSt, 3000 bbl at 100 cSt, 4000 bbl at 201 cSt (130 °F).

> 2019 Q3(b) 8 marks
>
> Same as 2022 4(a).

> 2019 Q3(c) 18 marks
>
> Same with 75, 100 and 200 cSt.

Full answer: ESE 3203 Study Pack, topic A7.

Section B: Natural gas processing

topics B1-B7 (usually questions 5-8; questions 1-4 in 2021)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B5 Gas sweetening: reasons for removing H_2S and CO_2, amine process, iron sponge, molecular sieve, sulfur recovery

Pass set · Very likely · 95% · Easy

Asked in 5 of 6 papers · 70 marks in total

STUDY THESE

- Acid gases: H_2S (toxic, fatal above 600 ppm, corrosive, burns to SO_2, poisons catalysts); CO_2 (no heating value, carbonic acid corrosion, freezes in LNG plants);

both promote hydrates

- Amine (MEA, DEA, MDEA): lean amine absorbs acid gas in a contactor (counter-current, ~38 °C, high P); rich amine flashed, heated, stripped in a regenerator (~115-125 °C, low P) and recycled
- Iron sponge: Fe_2O_3 on wood chips: $Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O$; batch, for < 300 ppm H_2S, does not remove CO_2; regenerated with air: $2Fe_2S_3 + 3O_2 \rightarrow 2Fe_2O_3 + 6S$
- Claus: $H_2S + 1.5O_2 \rightarrow SO_2 + H_2O$; $2H_2S + SO_2 \rightarrow 3S + 2H_2O$

> 2023 Q8(a) 12 marks
>
> Explain the alkaline amine process.

> 2023 Q7(b) 8 marks
>
> Reasons for removing H_2S and CO_2 from natural gas.

> 2022B Q7(b) 8 marks
>
> Short notes: desulphurization and regeneration.

> 2022 Q7(a) 12 marks
>
> Alkaline amine process for gas sweetening with a block diagram.

> 2021 Q3(c) 10 marks
>
> Iron sponge unit for removal of H_2S with a schematic.

> 2021 Q4(c) 10 marks
>
> Iron sponge vs solid bed (molecular sieve) sweetening for 3% CO_2, 0.02% H_2S on a low budget: which and why.

> 2020 Q4(c) 10 marks
>
> Same as 2023 7(b).

Full answer: ESE 3203 Study Pack, topic B5.

B6 LNG and LPG: LNG plant and refrigeration cycles, storage and carriers, LPG uses and hazards, LPG vs LNG vs CNG, NGL recovery

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 202 marks in total

STUDY THESE

- LNG: natural gas cooled to −162 °C at ~1 atm, 1/600 of gas volume; plant: inlet separation → acid gas removal → dehydration → mercury removal → NGL recovery → liquefaction (cascade, mixed refrigerant, C3-MR, expander) → N_2 rejection → storage
- Cascade: propane (−35 °C), ethylene (−100 °C), methane (−160 °C) cycles; each refrigerant has a limited range, so three are needed to span ambient to −162 °C with small temperature approach
- LPG: propane/butane mix (Bangladesh ~30/70), liquid at 4-15 bar, heavier than air, odorized; uses domestic, commercial, industrial, autogas, refrigerant, feedstock; hazards: leaks, fire/BLEVE, asphyxiation, cold burns
- LNG carriers: double hull, Moss spherical or membrane tanks (9% Ni steel, aluminium, Invar), insulation, boil-off used as fuel

> `2023` `Q7(a)` `12 marks`
>
> With a schematic of an LNG plant, describe the overall LNG production process.

> `2023` `Q8(b)` `8 marks`
>
> Commercial and industrial uses of LPG.

> `2022B` `Q7(a)` `12 marks`
>
> Same as 2023 7(a).

> `2022B` `Q8(d)` `10 marks`
>
> Same as 2021 8(b).

> `2022` `Q7(c)` `15 marks`
>
> What is LNG? Properties; schematic of an LNG plant and overall production process.

> `2022` `Q8(b)` `15 marks`
>
> Potential hazards and remedies of LPG.

> `2021` `Q7(a)` `10 marks`
>
> Construction of an LNG carrier with a figure.

> `2021` `Q7(b)` `10 marks`
>
> Same as 2023 8(b).

> `2021` `Q7(c)` `15 marks`
>
> Potential hazards and safety legislation of LPG.

> `2021` `Q8(b)` `12 marks`
>
> Tabulate major differences among LPG, LNG and CNG.

> 2020 Q4(b) 5 marks
>
> Short note on LPG and LNG.

> 2019 Q5(a) 20 marks
>
> Same as 2022 7(c).

> 2019 Q5(b) 6 marks
>
> Different types of LNG plants.

> 2019 Q5(c) 9 marks
>
> Why at least three refrigerants are needed to liquefy natural gas.

> 2019 Q6(a) 20 marks
>
> Cascade LNG process with schematic and T-s diagrams of the refrigerants.

> 2019 Q8(c) 10 marks
>
> LPG and NGL extraction process with a flow diagram.

> 2019 Q7(a) 13 marks
>
> Kettle-type shell-and-tube heat exchanger with a schematic.

Full answer: ESE 3203 Study Pack, topic B6.

B1 Natural gas: what it is, origin, classification, condensate, reservoirs, Bangladesh outlook

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 166 marks in total

STUDY THESE

- Natural gas: mixture of light hydrocarbons, mainly methane, with ethane, propane, butanes and N_2, CO_2, H_2S, He; colourless, odourless, lighter than air
- Origin: thermogenic (organic matter buried, heated and compressed: kerogen → oil → gas), biogenic (methanogenic bacteria at shallow depth), abiogenic (deep inorganic, minor theory); traps under cap rock
- Classification: associated (with oil) vs non-associated; dry (lean) vs wet (rich in C_2^+); sweet vs sour; conventional vs unconventional (shale gas, tight gas, coal-bed methane, hydrates)
- Condensate: C_5^+ liquids that drop out when the gas cools below its hydrocarbon dew point; separated in field separators, stabilized, split into naphtha, kerosene, diesel; retrograde condensation in the reservoir

> **2023 Q5(a) 14 marks**
>
> What is natural gas condensate? Flow diagram of separation of condensate from raw natural gas.

> **2023 Q5(b) 11 marks**
>
> Classify natural gas by chemical composition. Potential of unconventional gas reserves in Bangladesh.

> **2022B Q5(a) 10 marks**
>
> Same as 2023 5(a).

> **2022B Q5(c) 10 marks**
>
> Explain the origin of natural gas with a neat sketch.

> **2022 Q5(a) 12 marks**
>
> Same as 2023 5(a).

> **2022 Q5(b) 13 marks**
>
> Describe the most prevalent theory of natural gas formation with a neat sketch.

> **2021 Q1(b) 6 marks**
>
> Recommendations if an unconventional reservoir is found in Bangladesh, from an engineering perspective.

> **2021 Q1(c) 8 marks**
>
> Briefly describe the theories regarding the origin of natural gas.

> **2021 Q1(a) 5 marks**
>
> Is it better to burn natural gas or release it to the atmosphere without burning? Reasons.

> **2021 Q1(d) 16 marks**
>
> You have found a condensate gas reservoir: concerns about production and how to solve them.

> **2020 Q1(c) 10 marks**
>
> Same as 2023 5(a).

> **2020 Q5(a) 6 marks**
>
> “Natural gas is a clean burning fuel”: explain.

> **2020 Q5(b) 18 marks**
>
> Compare saturated black oil, undersaturated black oil and volatile oil reservoirs, with figures.

> **2019 Q7(b) 12 marks**

Phase envelope: bubble point line, dew point line, critical point, cricondenbar, cricondentherm, retrograde region.

2019 Q6(b) 15 marks

Short notes: gas quality aspect of LNG, Wobbe index, associated and non-associated gas.

Full answer: ESE 3203 Study Pack, topic B1.

B3 Gas-liquid separators: vertical, horizontal, double-barrel, spherical, three-phase; internals; sizing

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 135 marks in total

STUDY THESE

- Separation by momentum change (inlet diverter), gravity settling, coalescence (mist extractor); sections: primary (inlet), gravity settling, liquid collection, mist extractor
- Vertical: handles slugs and sand, small footprint, larger/costlier; horizontal: smaller and cheaper for high gas-oil ratio, larger interface, poor with solids and surges; double-barrel: liquid drains to lower barrel, no re-entrainment; spherical: compact, cheap, limited surge
- Internals: inlet diverter (baffle or centrifugal), wave breakers, defoaming plates, vortex breaker, mist extractor (wire mesh, vanes), level and pressure controllers
- Sizing (Souders-Brown): $D^2 = q_g Z(T + 460)\rho_g^{0.5}/[2.40Kp(\rho_L - \rho_g)^{0.5}]$; liquid V_L = Wt/1440; L/D 3-8

2023 Q6(a) 12 marks

Explain the working mechanism of a horizontal separator.

2023 Q6(b) 5 marks

Functions of de-foaming plates.

2023 Q6(c) 18 marks

Separator at 1000 psig, 7 MMscfd, GLR 40 bbl/MMscf: size vertical, horizontal single-tube and spherical separators (ρ_L 52 lbm/ft^3, γ_g 0.8, 110 °F, t = 3 min, 1/4 full).

2022B Q6(a) 15 marks

> Design and operation of a double-barrel horizontal separator with a figure.

> 2022B Q6(b) 10 marks
> Functions of de-foaming plates and vortex breaker.

> 2022 Q6(a) 15 marks
> Design and operation of a vertical separator with a figure.

> 2022 Q6(b) 10 marks
> Functions of centrifugal inlet diverter and wave breakers.

> 2021 Q3(b) 15 marks
> Gas production fell; which separator would you choose to replace a horizontal one to cut operating cost? Schematic; advantages and disadvantages.

> 2021 Q3(a) 10 marks
> What is vortex? Why is it undesirable in gas-liquid separation and how is it avoided?

> 2020 Q6(a) 10 marks
> Briefly describe a conventional gas-liquid separator with a schematic.

> 2019 Q8(a) 15 marks
> Schematic of a three-phase separator; its operation in petroleum production.

Full answer: ESE 3203 Study Pack, topic B3.

B4 Water in natural gas: dehydration, water content (Campbell), hydrates

Pass set · Very likely · 100% · Medium

Asked in 6 of 6 papers · 122 marks in total

STUDY THESE

- Why dehydrate: free water forms hydrates (plugging), causes corrosion with CO_2/H_2S, slugging, lower heating value; pipeline limit 6-8 lb/MMscf
- Water content falls with pressure, rises with temperature, falls with salinity and gas gravity; Campbell: $W = y_{HC}W_{HC} + y_{CO2}W_{CO2} + y_{H2S}W_{H2S}$ from three charts
- Hydrates: ice-like crystals of water cages holding gas molecules (CH_4, C_2H_6, C_3H_8, CO_2, H_2S); need free water, low T, high P, agitation; prevented by dehydration, heating, pressure reduction, methanol/glycol injection, kinetic and anti-agglomerant inhibitors

- Dehydration: absorption by glycol (TEG contactor + reboiler at 175-205 °C) or adsorption on solid desiccants (alumina, silica gel, molecular sieves: two towers, adsorb/regenerate)

2023 Q7(c) 15 marks

Water content of a gas at 1000 psia, 100 °F by Campbell's method: CH_4 80, C_2H_6 5, C_3H_8 1.5, nC_4 0.5, CO_2 2.5, N_2 2.0, H_2S 8.5 %.

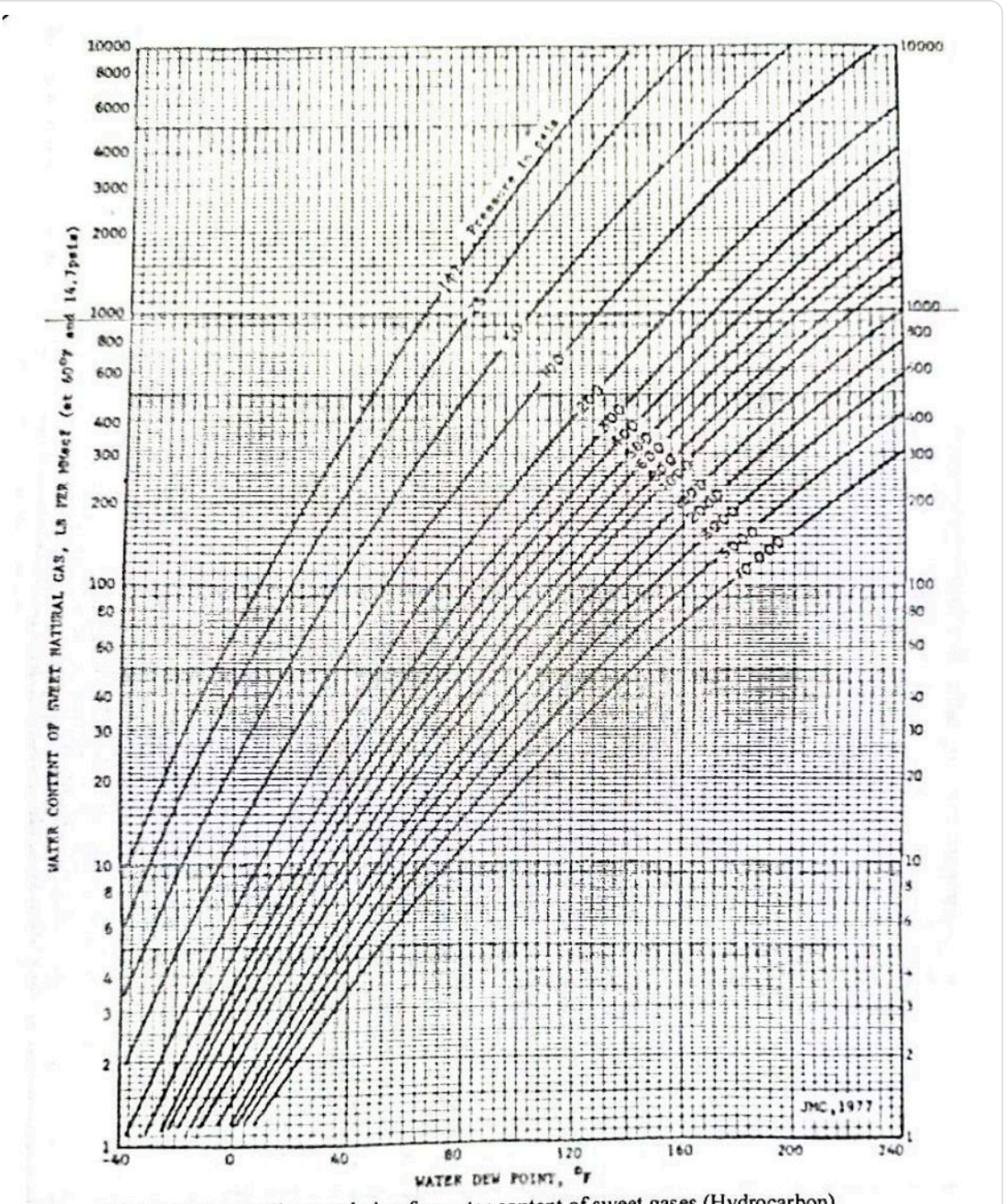


Figure 1: Campbell's correlation for water content of sweet gases (Hydrocarbon)

2022B Q5(b) 5 marks

Same as 2022 7(b).

2022B Q8(a) 8 marks

Reasons for removing water from natural gas; on what does water content depend?

2022B Q8(c) 10 marks

Same as 2023 7(c).

2022 Q7(b) 8 marks

Why is dehydration a must in natural gas processing?

2022 Q8(c) 10 marks

Same as 2023 7(c) (charts given).

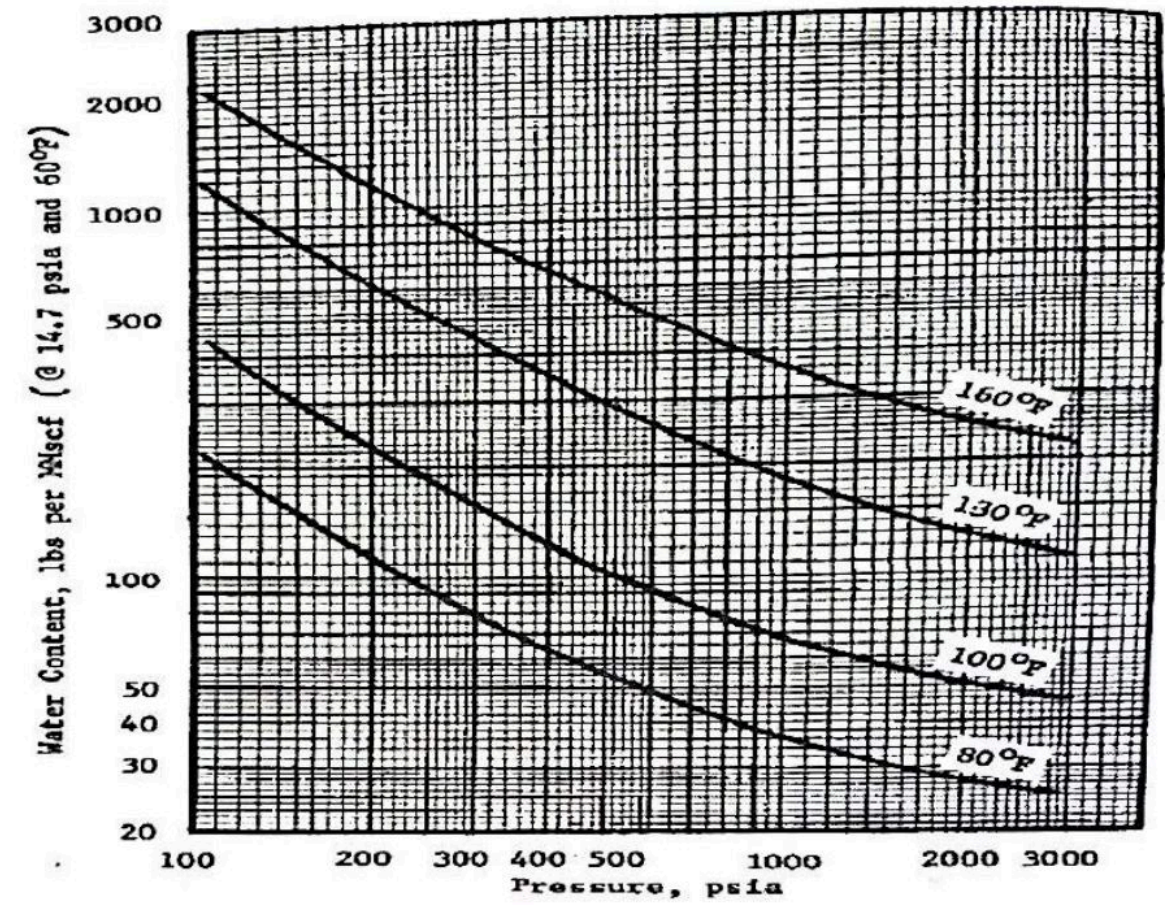


Figure 2: water content of CO_2 in saturated natural gas mixtures

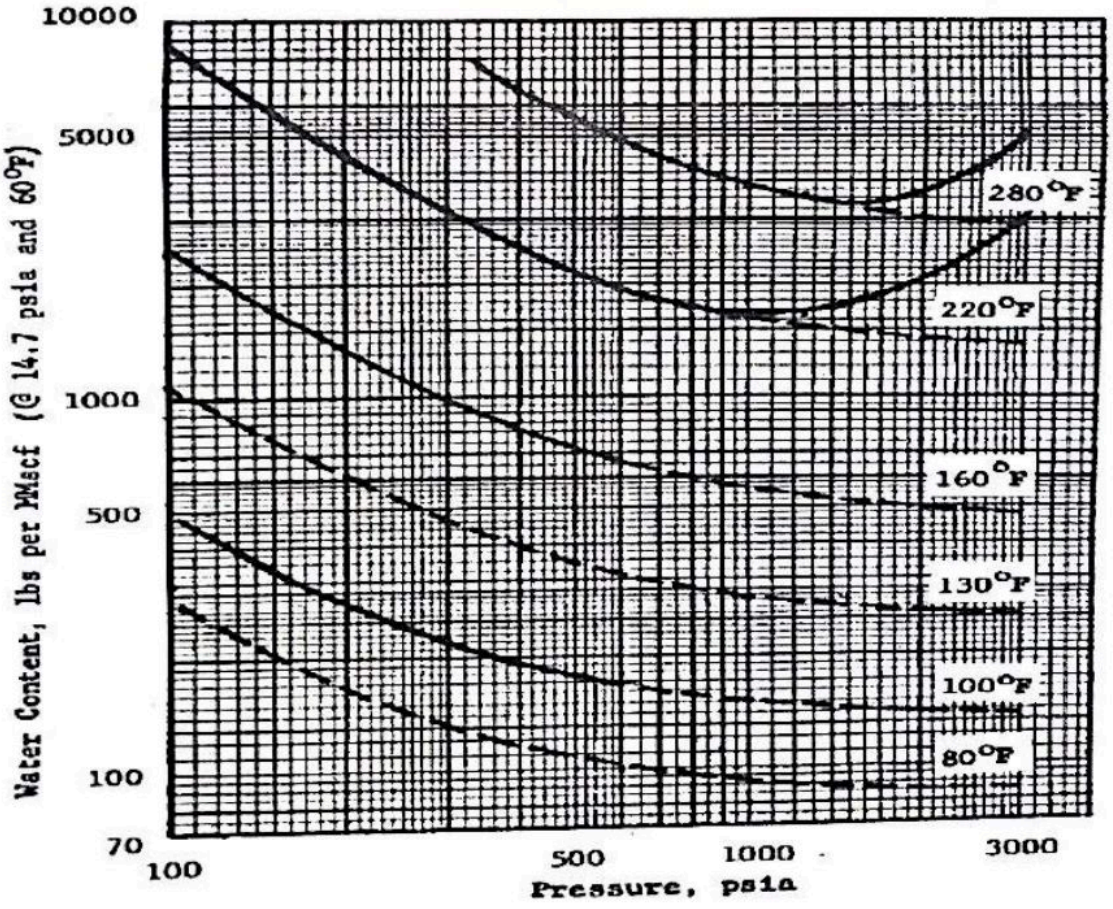


Figure 3: water content of H_2S in saturated natural gas mixtures

2021 Q4(a) 15 marks

Steps to avoid hydrate formation in pipelines, with the gas-water-hydrate equilibrium diagram.

2021 Q4(b) 10 marks

Alumina, silica gel, molecular sieve: which for sour gas, low cost, 12 ppm? Comparative analysis.

2020 Q5(c) 6 marks

Same as 2022 7(b).

2020 Q4(a) 15 marks

Explain natural gas hydrates with a diagram.

2019 Q7(c) 10 marks

Flow assurance in natural gas transport; what hydrates are and why they form.

2019 Q8(b) 10 marks

Natural gas dehydration by adsorption with a schematic.

Full answer: ESE 3203 Study Pack, topic B4.

B2 Gas properties and rock properties: z-factor, pseudo-reduced properties, density, B_g, compressibility, viscosity, porosity and permeability

Pass set · Very likely · 95% · Hard

Asked in 5 of 6 papers · 85 marks in total

STUDY THESE

- Real gas: $pV = znRT$; z from the Standing-Katz chart at $p_{pr} = p/p_{pc}$, $T_{pr} = T/T_{pc}$; Kay's rule $p_{pc} = \Sigma y_i p_{ci}$, $T_{pc} = \Sigma y_i T_{ci}$; Wichert-Aziz correction for CO_2, H_2S
- $\rho_g = pM_a/(zRT)$, R = 10.73 psia ft^3/(lb-mol °R); $B_g = 0.02827zT/p$ ft^3/scf; $c_g = 1/p - (1/z)(\partial z/\partial p)_T$
- Viscosity by Carr-Kobayashi-Burrows: μ_1 at 1 atm from M or γ_g and T, plus N_2, CO_2, H_2S corrections
- Porosity ϕ = pore volume/bulk volume (storage); permeability k = ability to transmit fluid (darcy, Darcy's law)

2023 Q5(c) 10 marks

> Elucidate: gas formation volume factor, isothermal compressibility of gases.

> 2022B Q5(d) 10 marks
>
> Porosity and permeability. Limestone: void 25 m^3, bulk 68 m^3: porosity.

> 2022 Q5(c) 10 marks
>
> Porosity vs permeability. Sandstone: void 35 cm^3, bulk 83 cm^3: porosity.

> 2021 Q2(a) 35 marks
>
> Gas C_1 80, C_2 3, C_3 2, nC_4 2, iC_4 1, nC_5 1, iC_5 3, nC_6 0.05, C_{7+} 1 (as nC_{10}), N_2 0.05, CO_2 2, H_2S 4 % at 180 °F, 1900 psia: z-factor and viscosity (no pressure correction).

> 2020 Q6(b) 20 marks
>
> Pseudo-reduced p and T for C_1 90%, C_2 3%, C_3 2%, H_2S 2%, N_2 3% at 200 °F, 2000 psia; density and B_g with z = 0.92.

Full answer: ESE 3203 Study Pack, topic B2.

B7 Reservoir flow and gas measurement: Darcy's law (linear and radial), orifice meter, choosing a meter

Next to add · Likely · 73% · Medium

Asked in 3 of 6 papers · 67 marks in total

STUDY THESE

- Darcy (field units, linear): $q = 0.001127kA(p_1 - p_2)/(\mu L)$ bbl/day with k in md, A ft^2, p psi, μ cp, L ft; apparent velocity v = q/A, actual (interstitial) velocity = v/ϕ
- Radial steady state: $p = p_{wf} + [Q_o B_o \mu_o/(0.00708kh)]\ln(r/r_w)$; most pressure drop is near the well
- Flow regimes: steady, unsteady (transient), pseudo-steady; geometries: radial, linear, spherical/hemispherical; fluids: incompressible, slightly compressible, compressible
- Orifice meter: thin plate with a hole; Δp across it gives flow by Bernoulli: $q = CA_o\sqrt{(2\Delta p/\rho)}$; AGA-3 for gas; selection by accuracy, range, pressure loss, cost, fluid cleanliness

> 2023 Q8(c) 15 marks
>
> Well at 600 STB/day, p_{wf} 1800 psi, k 120 md, h 25 ft, 40 acres, r_w 0.25 ft, B_o 1.25, μ_o 2.5 cp: pressure profile and drops across 1-ft intervals.

> **2022B Q6(c) 10 marks**
>
> Same as 2022 6(c).

> **2022B Q7(c) 15 marks**
>
> Same as 2023 8(c).

> **2022B Q8(b) 7 marks**
>
> Criteria for selecting a gas flow measurement device.

> **2022 Q6(c) 10 marks**
>
> Linear flow q = 0.001127kAΔp/(μL): L 2000 ft, h 20 ft, width 300 ft, k 100 md, ϕ 15%, μ 2 cp, p_1 2000, p_2 1990 psi: q, apparent and actual velocity.

> **2022 Q8(a) 10 marks**
>
> Explain the working principle of an orifice meter.

Full answer: ESE 3203 Study Pack, topic B7.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ADU	atmospheric distillation unit
AGA	American Gas Association
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
bbl	barrel (42 US gallons, 5.615 ft^3)
BLEVE	boiling liquid expanding vapour explosion
BPCD	barrels per calendar day
BPSD	barrels per stream day
CBM	coal-bed methane
CCR	continuous catalyst regeneration
CDU	crude distillation unit
CNG	compressed natural gas
cp	centipoise

cSt	centistokes
DEA	diethanolamine
EFV	equilibrium flash vaporization
ETBE	ethyl tertiary-butyl ether
FCC	fluid catalytic cracking
FSRU	floating storage and regasification unit
GLR	gas-liquid ratio
GOR	gas-oil ratio
HDN	hydrodenitrogenation
HDS	hydrodesulfurization
HHV	higher heating value
KHI	kinetic hydrate inhibitor
LNG	liquefied natural gas
LPG	liquefied petroleum gas
LSR	light straight-run (naphtha)
LTS	low-temperature separator
md	millidarcy
MDEA	methyldiethanolamine
MEA	monoethanolamine
MEK	methyl ethyl ketone
MMscfd	million standard cubic feet per day
MON	motor octane number
MTBE	methyl tertiary-butyl ether
NGC	natural gas condensate
NGL	natural gas liquids
NMP	N-methyl pyrrolidone
PON	posted (pump) octane number
PSA	pressure swing adsorption
psia	pounds per square inch absolute

psig	pounds per square inch gauge
RON	research octane number
RVP	Reid vapour pressure
scf	standard cubic foot
SR	semi-regenerative (reformer); also straight-run
STB	stock-tank barrel
TAME	tertiary-amyl methyl ether
TAN	total acid number
TBP	true boiling point
TEG	triethylene glycol
VDU	vacuum distillation unit
VI	viscosity index

Built from the scanned ESE 3203 papers (2019, 2020, 2021, 2022, 2022B, 2023). Question text is cleaned up from the scans.